deaths_gdp_analysis

May 13, 2026

1 Causes of Death by World Bank Region & Income Class (1980–2023)

[2]:
```
import pandas as pd
import numpy as np
import warnings
warnings.filterwarnings('ignore')

pd.set_option('display.max_columns', None)
pd.set_option('display.float_format', '{:.4f}'.format)
```

1.1 1. Load Raw Data

[3]:
```
world_raw = pd.read_csv('world.csv')

income_raw = pd.read_csv('incomeDeaths.csv')

gdpc_raw = pd.read_csv('gdpc.csv')

YEAR_START, YEAR_END = 1980, 2023

print('world.csv    :', world_raw.shape)
print('incomeDeaths.csv:', income_raw.shape)
print('gdpc.csv     :', gdpc_raw.shape)
```

```
world.csv    : (6468, 18)
incomeDeaths.csv: (3696, 11)
gdpc.csv     : (266, 70)
```

1.2 2. Cleaning

[4]:
```
year_cols = [str(y) for y in range(YEAR_START, YEAR_END + 1)]

gdpc_long = gdpc_raw.melt(
    id_vars=['Country Name', 'Country Code'],
    value_vars=year_cols,
    var_name='year',
```

```
    value_name='gdp_per_capita'
)
gdpc_long['year'] = gdpc_long['year'].astype(int)

region_gdp_map = {
    'East Asia & Pacific - WB'          : 'East Asia & Pacific',
    'Europe & Central Asia - WB'        : 'Europe & Central Asia',
    'Latin America & Caribbean - WB'    : 'Latin America & Caribbean',
    'Middle East & North Africa - WB'   : 'Middle East, North Africa,␣
↪Afghanistan & Pakistan',
    'North America'                     : 'North America',
    'South Asia - WB'                   : 'South Asia',
    'Sub-Saharan Africa - WB'           : 'Sub-Saharan Africa',
}

income_gdp_map = {
    'World Bank High Income'          : 'High income',
    'World Bank Low Income'           : 'Low income',
    'World Bank Lower Middle Income' : 'Lower middle income',
    'World Bank Upper Middle Income' : 'Upper middle income',
}

gdpc_indexed = (
    gdpc_long
    .set_index(['Country Name', 'year'])['gdp_per_capita']
)

print('GDP long shape:', gdpc_long.shape)
print('Sample lookup (East Asia & Pacific, 2010):',
      gdpc_indexed.get(('East Asia & Pacific', 2010), 'NOT FOUND'))
```

```
GDP long shape: (11704, 4)
Sample lookup (East Asia & Pacific, 2010): 7784.344658223
```

1.3 3. Build df_region

[5]:
```
region_deaths = world_raw[
    (world_raw['year'] >= YEAR_START) & (world_raw['year'] <= YEAR_END)
].copy()

region_deaths = region_deaths.rename(columns={
    'location_name' : 'region',
    'cause_name'    : 'cause',
    'metric_name'   : 'metric',
    'measure_name'  : 'measure',
    'val'           : 'death_share_pct',
    'upper'         : 'upper_pct',
```

```
    'lower'         : 'lower_pct',
})

region_deaths = region_deaths[[
    'region', 'cause', 'year', 'death_share_pct', 'upper_pct', 'lower_pct'
]]

region_deaths['gdp_country'] = region_deaths['region'].map(region_gdp_map)

region_deaths['gdp_per_capita_usd'] = region_deaths.apply(
    lambda r: gdpc_indexed.get((r['gdp_country'], r['year']), np.nan), axis=1
)

region_deaths = region_deaths.drop(columns='gdp_country')

df_region = region_deaths.sort_values(['region', 'cause', 'year']).
 ↪reset_index(drop=True)

print('df_region shape:', df_region.shape)
print('\nRegions:', df_region['region'].nunique(), '->',␣
 ↪sorted(df_region['region'].unique()))
print('Causes :', df_region['cause'].nunique())
print('Years  :', df_region['year'].min(), '-', df_region['year'].max())
print('\nGDP coverage (non-null):',
      df_region['gdp_per_capita_usd'].notna().sum(), '/', len(df_region))

df_region.head(10)
```

```
df_region shape: (6468, 7)

Regions: 7 -> ['East Asia & Pacific - WB', 'Europe & Central Asia - WB', 'Latin
America & Caribbean - WB', 'Middle East & North Africa - WB', 'North America',
'South Asia - WB', 'Sub-Saharan Africa - WB']
Causes : 21
Years  : 1980 - 2023

GDP coverage (non-null): 6468 / 6468
```

[5]:

```
                     region                    cause  year  death_share_pct  \
0  East Asia & Pacific - WB  Cardiovascular diseases  1980           0.3186
1  East Asia & Pacific - WB  Cardiovascular diseases  1981           0.3207
2  East Asia & Pacific - WB  Cardiovascular diseases  1982           0.3219
3  East Asia & Pacific - WB  Cardiovascular diseases  1983           0.3213
4  East Asia & Pacific - WB  Cardiovascular diseases  1984           0.3199
5  East Asia & Pacific - WB  Cardiovascular diseases  1985           0.3190
6  East Asia & Pacific - WB  Cardiovascular diseases  1986           0.3199
7  East Asia & Pacific - WB  Cardiovascular diseases  1987           0.3212
```

```
8  East Asia & Pacific - WB  Cardiovascular diseases  1988           0.3242
9  East Asia & Pacific - WB  Cardiovascular diseases  1989           0.3264

   upper_pct  lower_pct  gdp_per_capita_usd
0     0.3575     0.2703           1188.1785
1     0.3585     0.2742           1289.2773
2     0.3579     0.2786           1242.6616
3     0.3581     0.2800           1301.6217
4     0.3561     0.2775           1371.6239
5     0.3549     0.2787           1425.0739
6     0.3542     0.2805           1833.4874
7     0.3551     0.2832           2118.3190
8     0.3585     0.2843           2505.8672
9     0.3621     0.2848           2569.5527
```

1.4 4. Build df_income

[6]:
```python
income_deaths = income_raw[
    (income_raw['year'] >= YEAR_START) & (income_raw['year'] <= YEAR_END)
].copy()

income_deaths = income_deaths.rename(columns={
    'location' : 'income_class',
    'cause'    : 'cause',
    'val'      : 'death_share_pct',
    'upper'    : 'upper_pct',
    'lower'    : 'lower_pct',
})

income_deaths = income_deaths[[
    'income_class', 'cause', 'year', 'death_share_pct', 'upper_pct', 'lower_pct'
]]

income_deaths['gdp_country'] = income_deaths['income_class'].map(income_gdp_map)

income_deaths['gdp_per_capita_usd'] = income_deaths.apply(
    lambda r: gdpc_indexed.get((r['gdp_country'], r['year']), np.nan), axis=1
)

income_deaths = income_deaths.drop(columns='gdp_country')

df_income = income_deaths.sort_values(['income_class', 'cause', 'year']).
 ↪reset_index(drop=True)

print('df_income shape:', df_income.shape)
print('\nIncome classes:', df_income['income_class'].nunique(), '->',␣
 ↪sorted(df_income['income_class'].unique()))
```

```
print('Causes :', df_income['cause'].nunique())
print('Years  :', df_income['year'].min(), '-', df_income['year'].max())
print('\nGDP coverage (non-null):',
      df_income['gdp_per_capita_usd'].notna().sum(), '/', len(df_income))

df_income.sort_values("year").head(10)
```

```
df_income shape: (3696, 7)

Income classes: 4 -> ['World Bank High Income', 'World Bank Low Income', 'World
Bank Lower Middle Income', 'World Bank Upper Middle Income']
Causes : 21
Years  : 1980 - 2023

GDP coverage (non-null): 3696 / 3696
```

[6]:

```
                       income_class  \
1936  World Bank Lower Middle Income
836           World Bank High Income
3476  World Bank Upper Middle Income
440           World Bank High Income
1144           World Bank Low Income
1408           World Bank Low Income
2992  World Bank Upper Middle Income
3652  World Bank Upper Middle Income
704           World Bank High Income
528           World Bank High Income

                                         cause  year  death_share_pct  \
1936                 Diabetes and kidney diseases  1980           0.0183
836                           Transport injuries  1980           0.0269
3476          Self-harm and interpersonal violence  1980           0.0276
440                                    Neoplasms  1980           0.2155
1144  HIV/AIDS and sexually transmitted infections  1980           0.0068
1408                       Neurological disorders  1980           0.0068
2992  HIV/AIDS and sexually transmitted infections  1980           0.0015
3652                       Unintentional injuries  1980           0.0497
704           Self-harm and interpersonal violence  1980           0.0258
528                      Nutritional deficiencies  1980           0.0008

      upper_pct  lower_pct  gdp_per_capita_usd
1936     0.0219     0.0148            385.7892
836      0.0283     0.0257           8449.5621
3476     0.0333     0.0233            836.7354
440      0.2199     0.2082           8449.5621
1144     0.0121     0.0033            427.3882
1408     0.0117     0.0039            427.3882
```

```
2992      0.0023     0.0010           836.7354
3652      0.0563     0.0446           836.7354
704       0.0266     0.0251          8449.5621
528       0.0009     0.0007          8449.5621
```

[7]:
```
region_totals = (
    df_region.groupby(['region', 'year'])['death_share_pct']
    .sum()
    .reset_index()
    .rename(columns={'death_share_pct': 'total_pct'})
)
```

[8]:
```
income_totals = (
    df_income.groupby(['income_class', 'year'])['death_share_pct']
    .sum()
    .reset_index()
    .rename(columns={'death_share_pct': 'total_pct'})
)
```

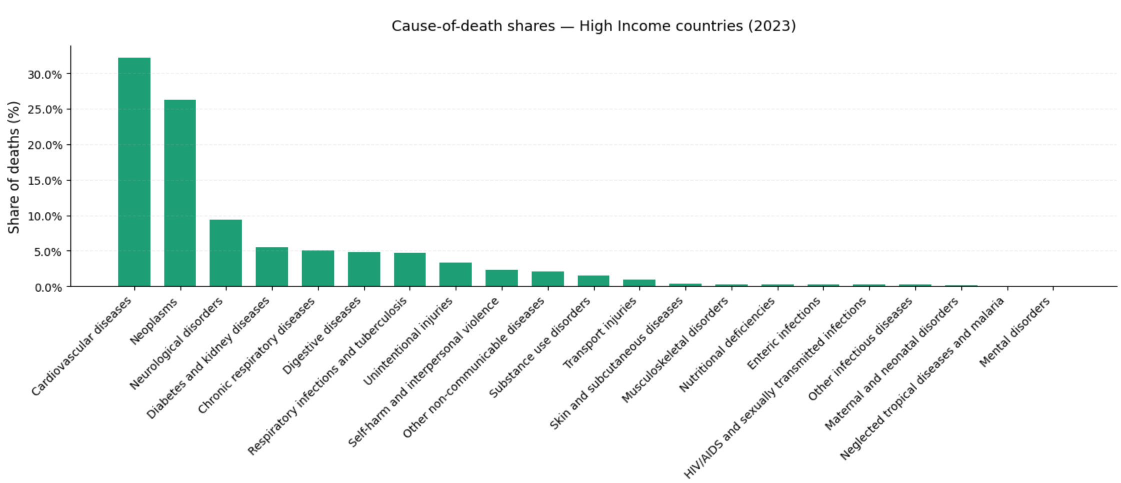
[9]:
```
sample_region = (
    df_region[(df_region['cause'] == 'Cardiovascular diseases') &␣
 ↪(df_region['year'] == 2010)]
    [['region', 'death_share_pct', 'gdp_per_capita_usd']]
    .sort_values('death_share_pct', ascending=False)
    .reset_index(drop=True)
)
print('Cardiovascular disease share by region (2010):')
print(sample_region.to_string())
```

```
Cardiovascular disease share by region (2010):
                            region  death_share_pct  gdp_per_capita_usd
0       Europe & Central Asia - WB           0.4620          23702.4750
1   Middle East & North Africa - WB           0.4209           4867.9294
2         East Asia & Pacific - WB           0.3605           7784.3447
3                    North America           0.3178          48545.7753
4                South Asia - WB           0.2477           1294.0273
5  Latin America & Caribbean - WB           0.2459           9183.9688
6         Sub-Saharan Africa - WB           0.1037           1642.6413
```

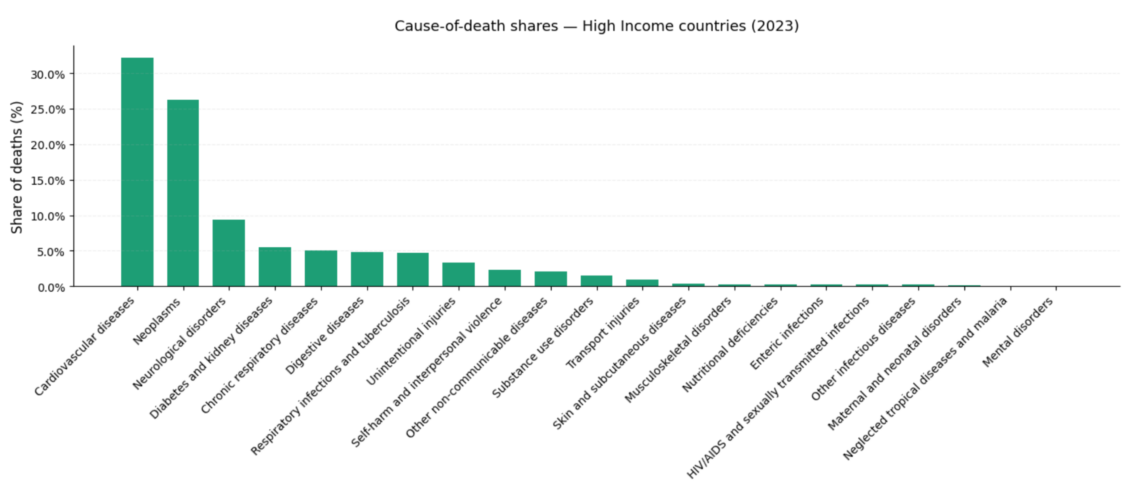
[10]:
```
sample_income = (
    df_income[(df_income['income_class'] == 'World Bank High Income') &␣
 ↪(df_income['year'] == 2023)]
    [['cause', 'death_share_pct', 'gdp_per_capita_usd']]
    .sort_values('death_share_pct', ascending=False)
    .reset_index(drop=True)
)
print('Cause-of-death shares for High Income countries (2023):')
print(sample_income.to_string())
```

```
Cause-of-death shares for High Income countries (2023):
                                           cause  death_share_pct
gdp_per_capita_usd
0                         Cardiovascular diseases           0.3222
48608.1948
1                                       Neoplasms           0.2629
48608.1948
2                          Neurological disorders           0.0935
48608.1948
3                    Diabetes and kidney diseases           0.0556
48608.1948
4                    Chronic respiratory diseases           0.0504
48608.1948
5                              Digestive diseases           0.0486
48608.1948
6        Respiratory infections and tuberculosis           0.0473
48608.1948
7                          Unintentional injuries           0.0328
48608.1948
8            Self-harm and interpersonal violence           0.0232
48608.1948
9                 Other non-communicable diseases           0.0211
48608.1948
10                       Substance use disorders           0.0151
48608.1948
11                            Transport injuries           0.0095
48608.1948
12                Skin and subcutaneous diseases           0.0032
48608.1948
13                    Musculoskeletal disorders           0.0031
48608.1948
14                     Nutritional deficiencies           0.0026
48608.1948
15                            Enteric infections           0.0024
48608.1948
16  HIV/AIDS and sexually transmitted infections           0.0022
48608.1948
17                    Other infectious diseases           0.0021
48608.1948
18               Maternal and neonatal disorders           0.0020
48608.1948
19       Neglected tropical diseases and malaria           0.0001
48608.1948
20                             Mental disorders           0.0000
48608.1948
```

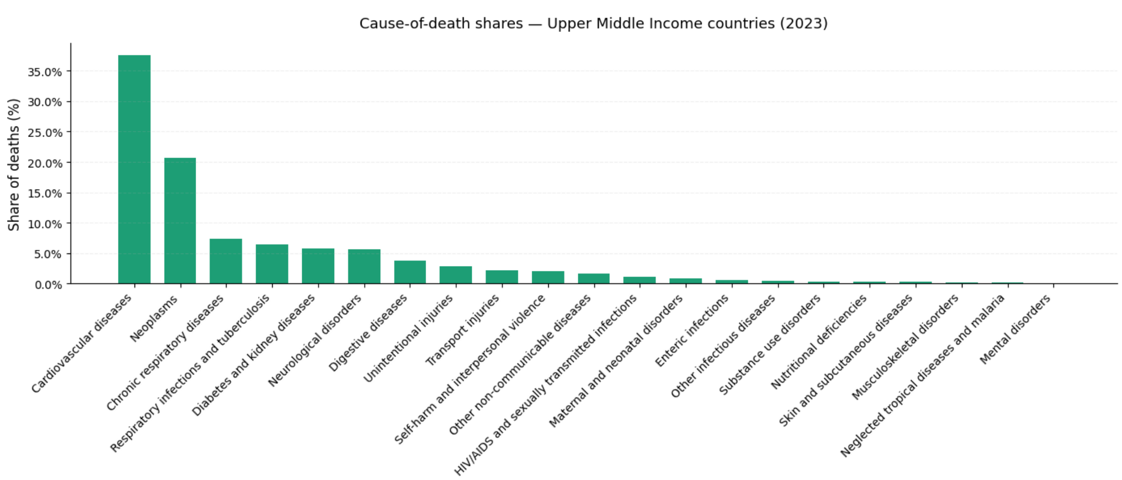
```
[11]: import matplotlib.pyplot as plt
import matplotlib.ticker as mticker

sample_income = (
    df_income[(df_income['income_class'] == 'World Bank High Income') &␣
  ↪(df_income['year'] == 2023)]
    [['cause', 'death_share_pct', 'gdp_per_capita_usd']]
    .sort_values('death_share_pct', ascending=False)
    .reset_index(drop=True)
)

fig, ax = plt.subplots(figsize=(14, 6))

causes = sample_income['cause']
shares = sample_income['death_share_pct'] * 100

ax.bar(causes, shares, color='#1D9E75', width=0.7)

ax.set_ylabel('Share of deaths (%)', fontsize=12)
ax.set_title('Cause-of-death shares — High Income countries (2023)',␣
  ↪fontsize=13, pad=14)
ax.yaxis.set_major_formatter(mticker.FormatStrFormatter('%.1f%%'))
ax.set_xticks(range(len(causes)))
ax.set_xticklabels(causes, rotation=45, ha='right', fontsize=10)
ax.spines[['top', 'right']].set_visible(False)
ax.grid(axis='y', alpha=0.2, linestyle='--')

plt.tight_layout()
plt.show()
```



[12]:
```python
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker

sample_income = (
    df_income[(df_income['income_class'] == 'World Bank High Income') & (df_income['year'] == 2023)]
    [['cause', 'death_share_pct', 'gdp_per_capita_usd']]
    .sort_values('death_share_pct', ascending=False)
    .reset_index(drop=True)
)

fig, ax = plt.subplots(figsize=(14, 6))

causes = sample_income['cause']
shares = sample_income['death_share_pct'] * 100

ax.bar(causes, shares, color='#1D9E75', width=0.7)

ax.set_ylabel('Share of deaths (%)', fontsize=12)
ax.set_title('Cause-of-death shares – High Income countries (2023)', fontsize=13, pad=14)
ax.yaxis.set_major_formatter(mticker.FormatStrFormatter('%.1f%%'))
ax.set_xticks(range(len(causes)))
ax.set_xticklabels(causes, rotation=45, ha='right', fontsize=10)
ax.spines[['top', 'right']].set_visible(False)
ax.grid(axis='y', alpha=0.2, linestyle='--')

plt.tight_layout()
plt.show()
```

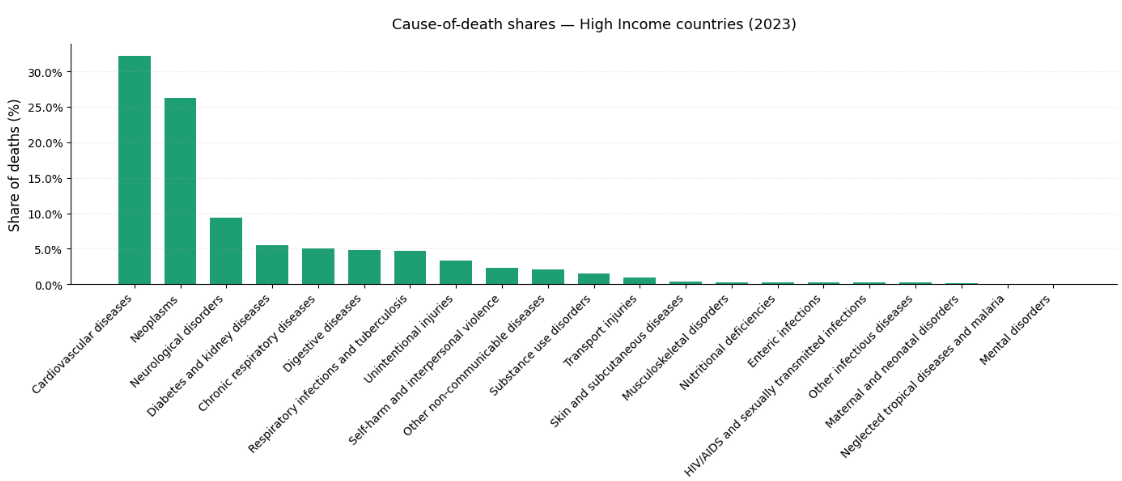


[13]:
```
sample_income = (
    df_income[(df_income['income_class'] == 'World Bank Upper Middle Income') & (df_income['year'] == 2023)]
    [['cause', 'death_share_pct', 'gdp_per_capita_usd']]
    .sort_values('death_share_pct', ascending=False)
    .reset_index(drop=True)
)

fig, ax = plt.subplots(figsize=(14, 6))

causes = sample_income['cause']
shares = sample_income['death_share_pct'] * 100

ax.bar(causes, shares, color='#1D9E75', width=0.7)

ax.set_ylabel('Share of deaths (%)', fontsize=12)
ax.set_title('Cause-of-death shares — Upper Middle Income countries (2023)', fontsize=13, pad=14)
ax.yaxis.set_major_formatter(mticker.FormatStrFormatter('%.1f%%'))
ax.set_xticks(range(len(causes)))
ax.set_xticklabels(causes, rotation=45, ha='right', fontsize=10)
ax.spines[['top', 'right']].set_visible(False)
ax.grid(axis='y', alpha=0.2, linestyle='--')

plt.tight_layout()
plt.show()
```



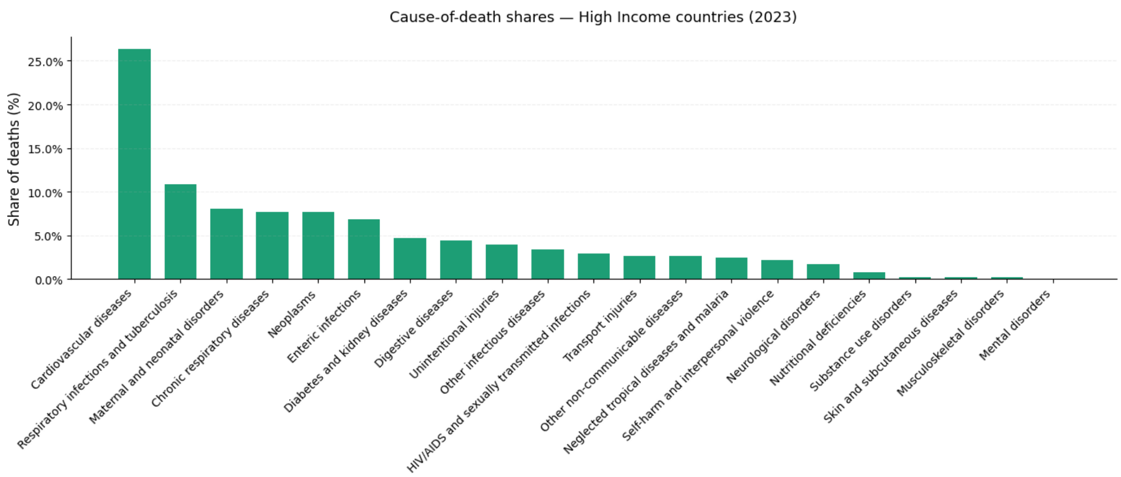
[14]:
```
sample_income = (
    df_income[(df_income['income_class'] == 'World Bank High Income') & (df_income['year'] == 2023)]
```

```
    [['cause', 'death_share_pct', 'gdp_per_capita_usd']]
    .sort_values('death_share_pct', ascending=False)
    .reset_index(drop=True)
)

fig, ax = plt.subplots(figsize=(14, 6))

causes = sample_income['cause']
shares = sample_income['death_share_pct'] * 100

ax.bar(causes, shares, color='#1D9E75', width=0.7)

ax.set_ylabel('Share of deaths (%)', fontsize=12)
ax.set_title('Cause-of-death shares — High Income countries (2023)',␣
↪fontsize=13, pad=14)
ax.yaxis.set_major_formatter(mticker.FormatStrFormatter('%.1f%%'))
ax.set_xticks(range(len(causes)))
ax.set_xticklabels(causes, rotation=45, ha='right', fontsize=10)
ax.spines[['top', 'right']].set_visible(False)
ax.grid(axis='y', alpha=0.2, linestyle='--')

plt.tight_layout()
plt.show()
```



[15]:
```
sample_income = (
    df_income[(df_income['income_class'] == 'World Bank Lower Middle Income') &␣
↪(df_income['year'] == 2013)]
    [['cause', 'death_share_pct', 'gdp_per_capita_usd']]
    .sort_values('death_share_pct', ascending=False)
    .reset_index(drop=True)
```

```
)

fig, ax = plt.subplots(figsize=(14, 6))

causes = sample_income['cause']
shares = sample_income['death_share_pct'] * 100

ax.bar(causes, shares, color='#1D9E75', width=0.7)

ax.set_ylabel('Share of deaths (%)', fontsize=12)
ax.set_title('Cause-of-death shares — High Income countries (2023)',␣
↪fontsize=13, pad=14)
ax.yaxis.set_major_formatter(mticker.FormatStrFormatter('%.1f%%'))
ax.set_xticks(range(len(causes)))
ax.set_xticklabels(causes, rotation=45, ha='right', fontsize=10)
ax.spines[['top', 'right']].set_visible(False)
ax.grid(axis='y', alpha=0.2, linestyle='--')

plt.tight_layout()
plt.show()
```



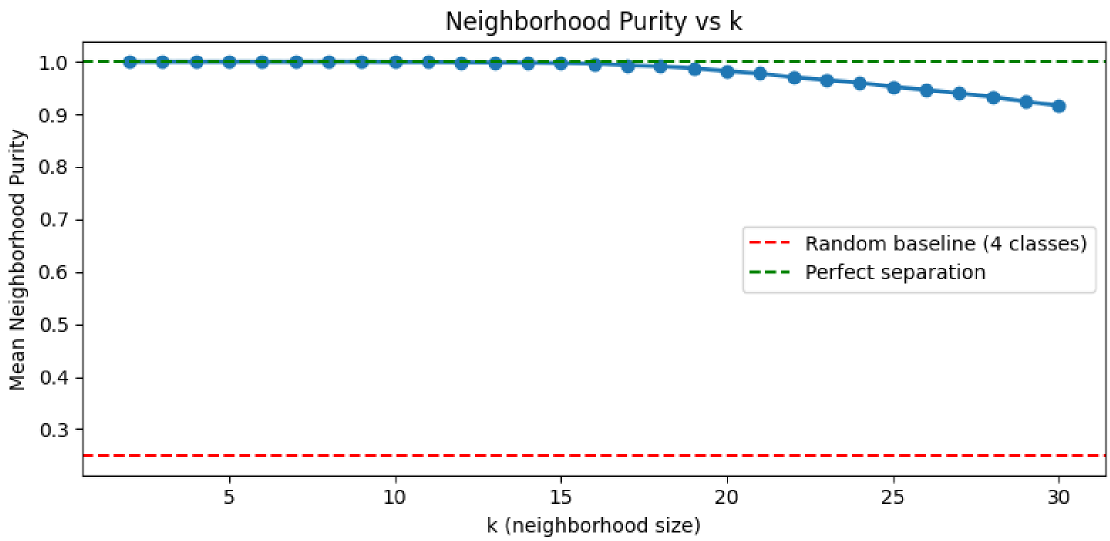
[30]:
```
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import silhouette_score, silhouette_samples
from sklearn.neighbors import NearestNeighbors
import matplotlib.pyplot as plt
import seaborn as sns

death_pivot = df_income.pivot_table(
```

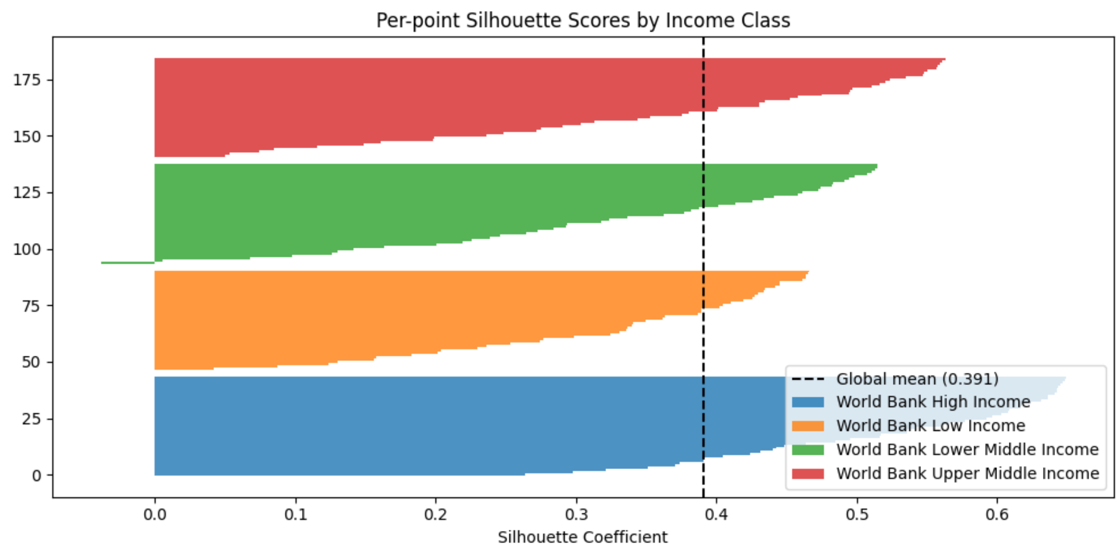
```
    index=['income_class', 'year'],
    columns='cause',
    values='death_share_pct'
).reset_index()

df_points = death_pivot.copy()

feature_cols = [c for c in df_points.columns if c not in ['income_class',␣
↪'year']]
X_raw = df_points[feature_cols].values
labels = df_points['income_class'].values

le = LabelEncoder()
y = le.fit_transform(labels)

X = StandardScaler().fit_transform(X_raw)

print(f"Feature matrix shape: {X.shape}")

sil_global = silhouette_score(X, y)
sil_samples = silhouette_samples(X, y)

print(f"\nGlobal Silhouette Score: {sil_global:.4f}")

sil_per_class = {}
for i, cls in enumerate(le.classes_):
    mask = y == i
    sil_per_class[cls] = sil_samples[mask].mean()
    print(f"  {cls:<40} {sil_per_class[cls]:.4f}")

def neighborhood_purity(X, y, k=10):

    nn = NearestNeighbors(n_neighbors=k + 1)
    nn.fit(X)
    distances, indices = nn.kneighbors(X)

    purities = []
    for i, neighbors in enumerate(indices):
        neighbors = neighbors[1:]
        same_class = np.sum(y[neighbors] == y[i])
        purities.append(same_class / k)

    return np.array(purities)

K = 10
purity_per_point = neighborhood_purity(X, y, k=K)
print(f"\nNeighborhood Purity (k={K})")
```

```
print(f"  Global mean: {purity_per_point.mean():.4f}")

purity_per_class = {}
for i, cls in enumerate(le.classes_):
    mask = y == i
    purity_per_class[cls] = purity_per_point[mask].mean()
    print(f"  {cls:<40} {purity_per_class[cls]:.4f}")

k_values = range(2, 31)
mean_purities = [neighborhood_purity(X, y, k=k).mean() for k in k_values]

plt.figure(figsize=(8, 4))
plt.plot(k_values, mean_purities, marker='o', linewidth=2)
plt.axhline(0.25, color='red', linestyle='--', label='Random baseline (4␣
 ↪classes)')
plt.axhline(1.0,  color='green', linestyle='--', label='Perfect separation')
plt.xlabel('k (neighborhood size)')
plt.ylabel('Mean Neighborhood Purity')
plt.title('Neighborhood Purity vs k')
plt.legend()
plt.tight_layout()
plt.show()

df_sil = pd.DataFrame({
    'income_class': labels,
    'silhouette':   sil_samples,
    'year':         df_points['year'].values
})

fig, ax = plt.subplots(figsize=(10, 5))
colors = sns.color_palette('tab10', n_colors=4)
y_lower = 0
for i, cls in enumerate(le.classes_):
    vals = np.sort(df_sil[df_sil['income_class'] == cls]['silhouette'].values)
    ax.barh(range(y_lower, y_lower + len(vals)), vals, height=1,
            color=colors[i], alpha=0.8, label=cls)
    y_lower += len(vals) + 3

ax.axvline(sil_global, color='black', linestyle='--', label=f'Global mean␣
 ↪({sil_global:.3f})')
ax.set_xlabel('Silhouette Coefficient')
ax.set_title('Per-point Silhouette Scores by Income Class')
ax.legend(loc='lower right')
plt.tight_layout()
plt.show()
```

```
Feature matrix shape: (176, 21)
```

```
Global Silhouette Score: 0.3910
  World Bank High Income                  0.5244
  World Bank Low Income                   0.3305
  World Bank Lower Middle Income          0.3343
  World Bank Upper Middle Income          0.3748

Neighborhood Purity (k=10)
  Global mean: 0.9994
  World Bank High Income                  1.0000
  World Bank Low Income                   1.0000
  World Bank Lower Middle Income          1.0000
  World Bank Upper Middle Income          0.9977
```





[17]:

```
death_pivot.head(44)
```

[17]:

```
cause            income_class  year  Cardiovascular diseases  \
0      World Bank High Income  1980                   0.4646
1      World Bank High Income  1981                   0.4629
2      World Bank High Income  1982                   0.4611
3      World Bank High Income  1983                   0.4611
4      World Bank High Income  1984                   0.4607
5      World Bank High Income  1985                   0.4595
6      World Bank High Income  1986                   0.4557
7      World Bank High Income  1987                   0.4528
8      World Bank High Income  1988                   0.4490
9      World Bank High Income  1989                   0.4424
10     World Bank High Income  1990                   0.4387
11     World Bank High Income  1991                   0.4365
12     World Bank High Income  1992                   0.4337
13     World Bank High Income  1993                   0.4351
14     World Bank High Income  1994                   0.4332
15     World Bank High Income  1995                   0.4284
16     World Bank High Income  1996                   0.4266
17     World Bank High Income  1997                   0.4248
18     World Bank High Income  1998                   0.4221
19     World Bank High Income  1999                   0.4204
20     World Bank High Income  2000                   0.4165
21     World Bank High Income  2001                   0.4145
22     World Bank High Income  2002                   0.4137
23     World Bank High Income  2003                   0.4107
24     World Bank High Income  2004                   0.4041
25     World Bank High Income  2005                   0.4006
26     World Bank High Income  2006                   0.3951
27     World Bank High Income  2007                   0.3896
28     World Bank High Income  2008                   0.3855
29     World Bank High Income  2009                   0.3808
30     World Bank High Income  2010                   0.3780
31     World Bank High Income  2011                   0.3700
32     World Bank High Income  2012                   0.3668
33     World Bank High Income  2013                   0.3617
34     World Bank High Income  2014                   0.3565
35     World Bank High Income  2015                   0.3545
36     World Bank High Income  2016                   0.3495
37     World Bank High Income  2017                   0.3467
38     World Bank High Income  2018                   0.3437
39     World Bank High Income  2019                   0.3391
40     World Bank High Income  2020                   0.3083
```

41	World Bank High Income	2021	0.2932
42	World Bank High Income	2022	0.3069
43	World Bank High Income	2023	0.3222

cause	Chronic respiratory diseases	Diabetes and kidney diseases \
0	0.0417	0.0273
1	0.0420	0.0275
2	0.0425	0.0277
3	0.0432	0.0280
4	0.0435	0.0282
5	0.0436	0.0284
6	0.0435	0.0288
7	0.0431	0.0289
8	0.0433	0.0294
9	0.0440	0.0301
10	0.0442	0.0309
11	0.0441	0.0313
12	0.0440	0.0314
13	0.0440	0.0312
14	0.0436	0.0312
15	0.0438	0.0319
16	0.0443	0.0326
17	0.0448	0.0332
18	0.0453	0.0339
19	0.0464	0.0347
20	0.0459	0.0361
21	0.0452	0.0372
22	0.0453	0.0379
23	0.0456	0.0387
24	0.0452	0.0392
25	0.0458	0.0400
26	0.0457	0.0407
27	0.0465	0.0414
28	0.0472	0.0418
29	0.0477	0.0422
30	0.0479	0.0430
31	0.0490	0.0439
32	0.0498	0.0449
33	0.0504	0.0457
34	0.0506	0.0467
35	0.0517	0.0482
36	0.0520	0.0494
37	0.0528	0.0509
38	0.0528	0.0519
39	0.0526	0.0533
40	0.0445	0.0496
41	0.0422	0.0480

42	0.0464	0.0515
43	0.0504	0.0556

cause	Digestive diseases	Enteric infections \
0	0.0430	0.0022
1	0.0428	0.0020
2	0.0424	0.0019
3	0.0421	0.0018
4	0.0419	0.0017
5	0.0418	0.0016
6	0.0414	0.0015
7	0.0410	0.0014
8	0.0409	0.0013
9	0.0413	0.0013
10	0.0411	0.0013
11	0.0411	0.0012
12	0.0411	0.0012
13	0.0408	0.0012
14	0.0411	0.0012
15	0.0413	0.0011
16	0.0414	0.0011
17	0.0415	0.0011
18	0.0416	0.0011
19	0.0417	0.0012
20	0.0423	0.0013
21	0.0430	0.0014
22	0.0435	0.0015
23	0.0441	0.0017
24	0.0448	0.0019
25	0.0452	0.0021
26	0.0457	0.0024
27	0.0461	0.0027
28	0.0465	0.0028
29	0.0468	0.0028
30	0.0470	0.0028
31	0.0472	0.0028
32	0.0472	0.0028
33	0.0474	0.0028
34	0.0478	0.0027
35	0.0479	0.0027
36	0.0480	0.0026
37	0.0478	0.0026
38	0.0480	0.0025
39	0.0485	0.0025
40	0.0444	0.0021
41	0.0432	0.0021
42	0.0461	0.0023

43	0.0486	0.0024

cause	HIV/AIDS and sexually transmitted infections \
0	0.0008
1	0.0008
2	0.0008
3	0.0008
4	0.0010
5	0.0012
6	0.0016
7	0.0022
8	0.0028
9	0.0036
10	0.0042
11	0.0049
12	0.0056
13	0.0060
14	0.0064
15	0.0062
16	0.0053
17	0.0042
18	0.0036
19	0.0034
20	0.0033
21	0.0033
22	0.0032
23	0.0031
24	0.0031
25	0.0031
26	0.0031
27	0.0030
28	0.0029
29	0.0029
30	0.0028
31	0.0028
32	0.0028
33	0.0029
34	0.0030
35	0.0030
36	0.0030
37	0.0030
38	0.0029
39	0.0028
40	0.0025
41	0.0022
42	0.0022
43	0.0022

cause	Maternal and neonatal disorders	Mental disorders	\
0	0.0137	0.0000	
1	0.0130	0.0000	
2	0.0123	0.0000	
3	0.0114	0.0000	
4	0.0108	0.0000	
5	0.0103	0.0000	
6	0.0099	0.0000	
7	0.0095	0.0000	
8	0.0091	0.0000	
9	0.0086	0.0000	
10	0.0081	0.0000	
11	0.0075	0.0000	
12	0.0069	0.0000	
13	0.0063	0.0000	
14	0.0059	0.0000	
15	0.0055	0.0000	
16	0.0052	0.0000	
17	0.0050	0.0000	
18	0.0049	0.0000	
19	0.0046	0.0000	
20	0.0045	0.0000	
21	0.0044	0.0000	
22	0.0042	0.0000	
23	0.0041	0.0000	
24	0.0041	0.0000	
25	0.0039	0.0000	
26	0.0040	0.0000	
27	0.0039	0.0000	
28	0.0037	0.0000	
29	0.0036	0.0000	
30	0.0034	0.0000	
31	0.0033	0.0000	
32	0.0033	0.0000	
33	0.0032	0.0000	
34	0.0031	0.0000	
35	0.0030	0.0000	
36	0.0029	0.0000	
37	0.0027	0.0000	
38	0.0025	0.0000	
39	0.0024	0.0000	
40	0.0020	0.0000	
41	0.0020	0.0000	
42	0.0020	0.0000	
43	0.0020	0.0000	

cause	Musculoskeletal disorders	Neglected tropical diseases and malaria	\
0	0.0021	0.0002	
1	0.0021	0.0002	
2	0.0021	0.0002	
3	0.0021	0.0002	
4	0.0022	0.0002	
5	0.0022	0.0002	
6	0.0023	0.0002	
7	0.0023	0.0001	
8	0.0023	0.0001	
9	0.0023	0.0001	
10	0.0024	0.0001	
11	0.0024	0.0001	
12	0.0024	0.0001	
13	0.0023	0.0001	
14	0.0024	0.0001	
15	0.0025	0.0001	
16	0.0025	0.0001	
17	0.0026	0.0001	
18	0.0026	0.0001	
19	0.0026	0.0001	
20	0.0027	0.0001	
21	0.0028	0.0001	
22	0.0029	0.0001	
23	0.0028	0.0001	
24	0.0028	0.0001	
25	0.0028	0.0001	
26	0.0028	0.0001	
27	0.0028	0.0001	
28	0.0028	0.0001	
29	0.0028	0.0001	
30	0.0028	0.0001	
31	0.0028	0.0001	
32	0.0028	0.0001	
33	0.0029	0.0001	
34	0.0029	0.0001	
35	0.0029	0.0001	
36	0.0030	0.0001	
37	0.0030	0.0001	
38	0.0030	0.0001	
39	0.0030	0.0001	
40	0.0027	0.0001	
41	0.0026	0.0001	
42	0.0029	0.0001	
43	0.0031	0.0001	

cause	Neoplasms	Neurological disorders	Nutritional deficiencies	\

0	0.2155	0.0349	0.0008
1	0.2184	0.0360	0.0008
2	0.2217	0.0369	0.0008
3	0.2230	0.0378	0.0008
4	0.2259	0.0387	0.0008
5	0.2297	0.0398	0.0009
6	0.2336	0.0413	0.0009
7	0.2371	0.0424	0.0009
8	0.2392	0.0434	0.0009
9	0.2411	0.0444	0.0009
10	0.2430	0.0452	0.0009
11	0.2443	0.0461	0.0009
12	0.2446	0.0467	0.0009
13	0.2401	0.0465	0.0009
14	0.2385	0.0469	0.0009
15	0.2413	0.0485	0.0010
16	0.2447	0.0502	0.0010
17	0.2473	0.0518	0.0011
18	0.2488	0.0529	0.0011
19	0.2471	0.0537	0.0011
20	0.2487	0.0548	0.0011
21	0.2494	0.0560	0.0011
22	0.2485	0.0570	0.0011
23	0.2486	0.0583	0.0011
24	0.2526	0.0596	0.0011
25	0.2525	0.0616	0.0011
26	0.2569	0.0638	0.0011
27	0.2598	0.0660	0.0011
28	0.2616	0.0679	0.0011
29	0.2645	0.0697	0.0011
30	0.2650	0.0718	0.0012
31	0.2675	0.0740	0.0012
32	0.2686	0.0764	0.0013
33	0.2700	0.0783	0.0013
34	0.2713	0.0800	0.0014
35	0.2683	0.0818	0.0015
36	0.2695	0.0835	0.0017
37	0.2680	0.0858	0.0019
38	0.2679	0.0874	0.0020
39	0.2687	0.0894	0.0022
40	0.2418	0.0815	0.0022
41	0.2308	0.0793	0.0023
42	0.2464	0.0858	0.0025
43	0.2629	0.0935	0.0026

cause	Other infectious diseases	Other non-communicable diseases	\
0	0.0040	0.0165	

1	0.0039	0.0164
2	0.0038	0.0164
3	0.0036	0.0162
4	0.0036	0.0162
5	0.0035	0.0162
6	0.0035	0.0163
7	0.0034	0.0163
8	0.0032	0.0161
9	0.0032	0.0160
10	0.0030	0.0156
11	0.0029	0.0152
12	0.0028	0.0148
13	0.0027	0.0141
14	0.0027	0.0139
15	0.0026	0.0139
16	0.0026	0.0139
17	0.0025	0.0139
18	0.0024	0.0140
19	0.0023	0.0140
20	0.0023	0.0141
21	0.0022	0.0142
22	0.0021	0.0142
23	0.0021	0.0143
24	0.0021	0.0146
25	0.0020	0.0149
26	0.0020	0.0153
27	0.0020	0.0156
28	0.0020	0.0158
29	0.0020	0.0160
30	0.0020	0.0163
31	0.0020	0.0165
32	0.0020	0.0168
33	0.0020	0.0171
34	0.0020	0.0177
35	0.0020	0.0181
36	0.0020	0.0186
37	0.0020	0.0189
38	0.0021	0.0192
39	0.0021	0.0197
40	0.0019	0.0183
41	0.0019	0.0180
42	0.0020	0.0195
43	0.0021	0.0211

cause	Respiratory infections and tuberculosis \
0	0.0399
1	0.0394

2	0.0388
3	0.0386
4	0.0362
5	0.0361
6	0.0362
7	0.0352
8	0.0356
9	0.0360
10	0.0360
11	0.0352
12	0.0355
13	0.0375
14	0.0380
15	0.0388
16	0.0391
17	0.0398
18	0.0398
19	0.0396
20	0.0382
21	0.0360
22	0.0361
23	0.0366
24	0.0362
25	0.0372
26	0.0363
27	0.0363
28	0.0369
29	0.0368
30	0.0366
31	0.0374
32	0.0378
33	0.0382
34	0.0382
35	0.0387
36	0.0383
37	0.0384
38	0.0387
39	0.0377
40	0.1282
41	0.1638
42	0.1043
43	0.0473

cause	Self-harm and interpersonal violence	Skin and subcutaneous diseases	\
0	0.0258	0.0009	
1	0.0260	0.0009	
2	0.0260	0.0009	

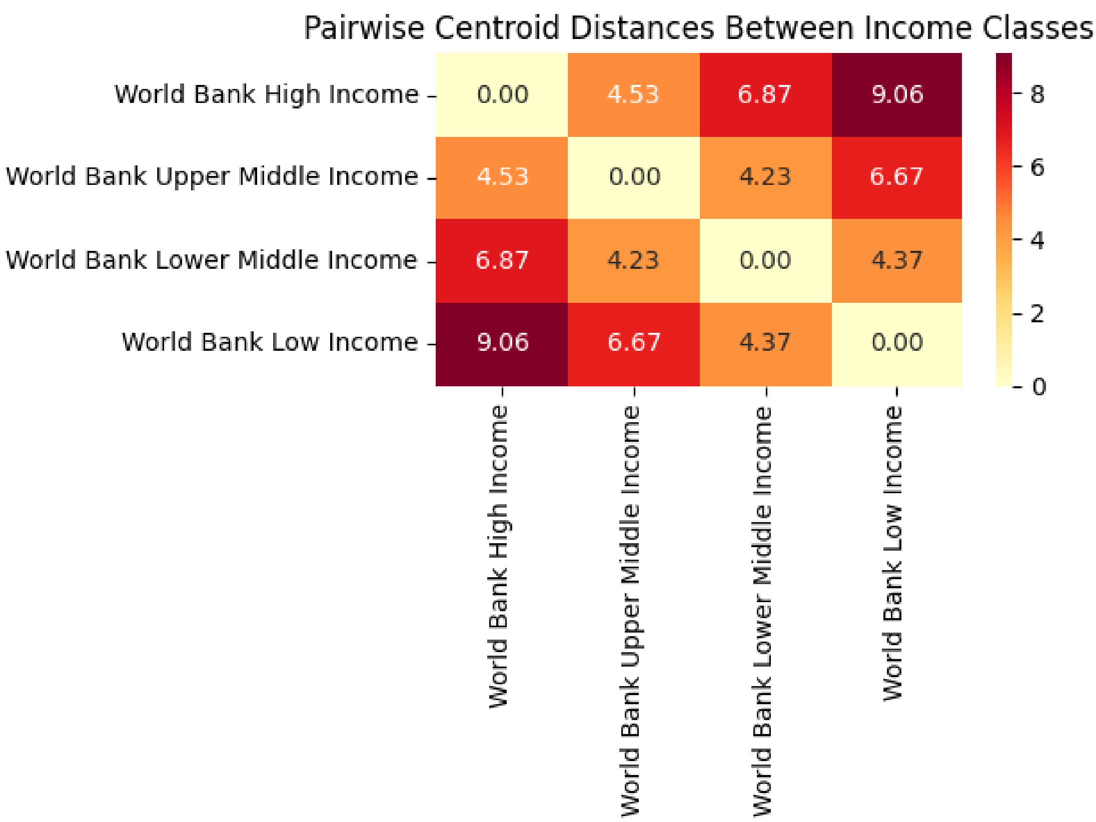
3	0.0260	0.0010
4	0.0261	0.0010
5	0.0253	0.0010
6	0.0249	0.0011
7	0.0247	0.0011
8	0.0246	0.0011
9	0.0253	0.0011
10	0.0257	0.0012
11	0.0267	0.0012
12	0.0279	0.0012
13	0.0292	0.0012
14	0.0303	0.0013
15	0.0301	0.0013
16	0.0289	0.0014
17	0.0279	0.0014
18	0.0278	0.0014
19	0.0287	0.0015
20	0.0289	0.0016
21	0.0289	0.0017
22	0.0282	0.0017
23	0.0277	0.0018
24	0.0278	0.0018
25	0.0270	0.0019
26	0.0261	0.0019
27	0.0255	0.0019
28	0.0251	0.0020
29	0.0249	0.0020
30	0.0244	0.0021
31	0.0240	0.0021
32	0.0235	0.0022
33	0.0230	0.0022
34	0.0227	0.0023
35	0.0219	0.0025
36	0.0216	0.0026
37	0.0208	0.0027
38	0.0203	0.0028
39	0.0202	0.0029
40	0.0184	0.0027
41	0.0178	0.0026
42	0.0210	0.0030
43	0.0232	0.0032

cause	Substance use disorders	Transport injuries	Unintentional injuries
0	0.0071	0.0269	0.0323
1	0.0071	0.0262	0.0316
2	0.0070	0.0255	0.0312
3	0.0070	0.0246	0.0307

4	0.0067	0.0244	0.0302
5	0.0058	0.0239	0.0292
6	0.0055	0.0239	0.0282
7	0.0055	0.0239	0.0280
8	0.0058	0.0240	0.0277
9	0.0061	0.0244	0.0279
10	0.0065	0.0241	0.0279
11	0.0069	0.0236	0.0280
12	0.0080	0.0228	0.0285
13	0.0097	0.0216	0.0295
14	0.0110	0.0212	0.0302
15	0.0107	0.0205	0.0305
16	0.0099	0.0199	0.0293
17	0.0092	0.0194	0.0283
18	0.0091	0.0192	0.0281
19	0.0095	0.0189	0.0283
20	0.0103	0.0188	0.0285
21	0.0108	0.0186	0.0291
22	0.0112	0.0182	0.0292
23	0.0115	0.0177	0.0295
24	0.0119	0.0176	0.0294
25	0.0119	0.0170	0.0292
26	0.0117	0.0168	0.0286
27	0.0112	0.0163	0.0281
28	0.0110	0.0154	0.0279
29	0.0108	0.0146	0.0279
30	0.0106	0.0138	0.0285
31	0.0104	0.0136	0.0293
32	0.0103	0.0131	0.0276
33	0.0106	0.0126	0.0276
34	0.0111	0.0124	0.0277
35	0.0114	0.0119	0.0280
36	0.0119	0.0116	0.0282
37	0.0122	0.0111	0.0286
38	0.0125	0.0107	0.0288
39	0.0131	0.0104	0.0291
40	0.0130	0.0092	0.0267
41	0.0134	0.0089	0.0256
42	0.0144	0.0092	0.0315
43	0.0151	0.0095	0.0328

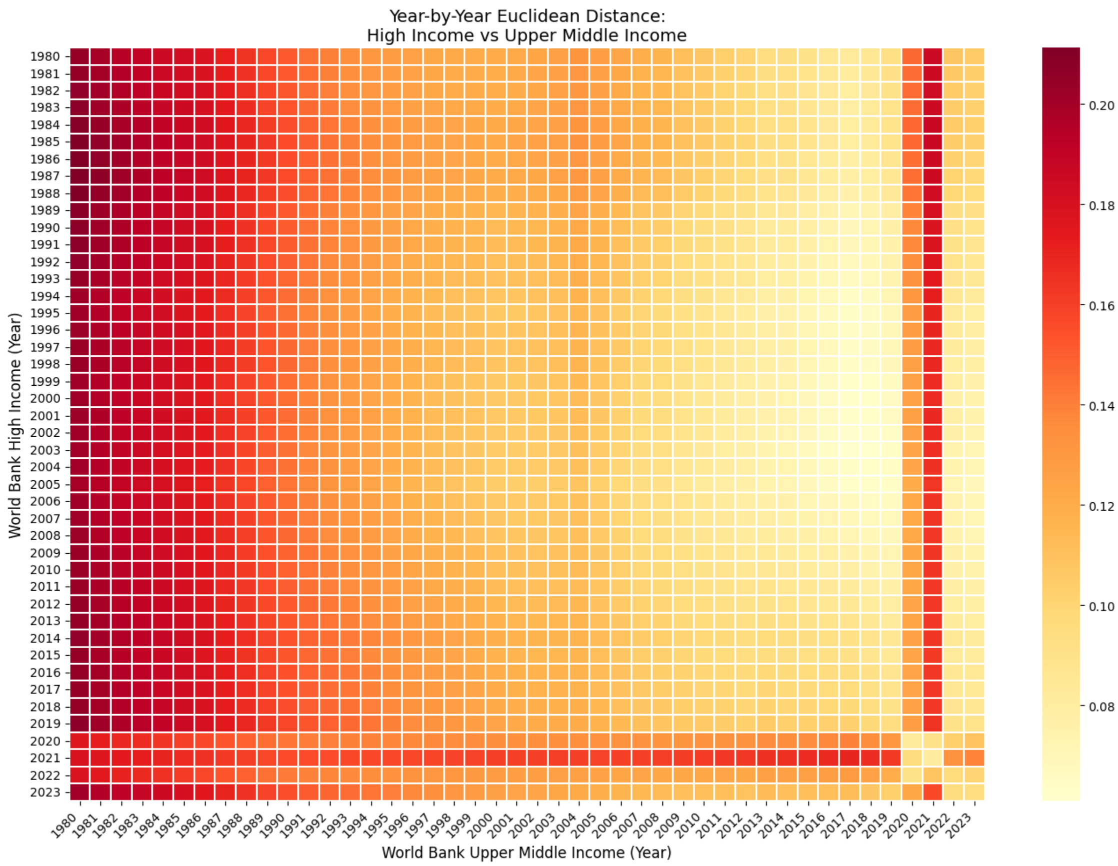
[31]:
```
from sklearn.metrics import pairwise_distances

centroids = np.array([X[y == i].mean(axis=0) for i in range(len(le.classes_))])

dist_matrix = pairwise_distances(centroids)
```

```python
df_dist = pd.DataFrame(dist_matrix, index=le.classes_, columns=le.classes_)

order = [
    'World Bank High Income',
    'World Bank Upper Middle Income',
    'World Bank Lower Middle Income',
    'World Bank Low Income'
]

df_dist_ordered = df_dist.loc[order, order]

sns.heatmap(df_dist_ordered, annot=True, fmt='.2f', cmap='YlOrRd')
plt.title('Pairwise Centroid Distances Between Income Classes')
plt.tight_layout()
plt.show()
```



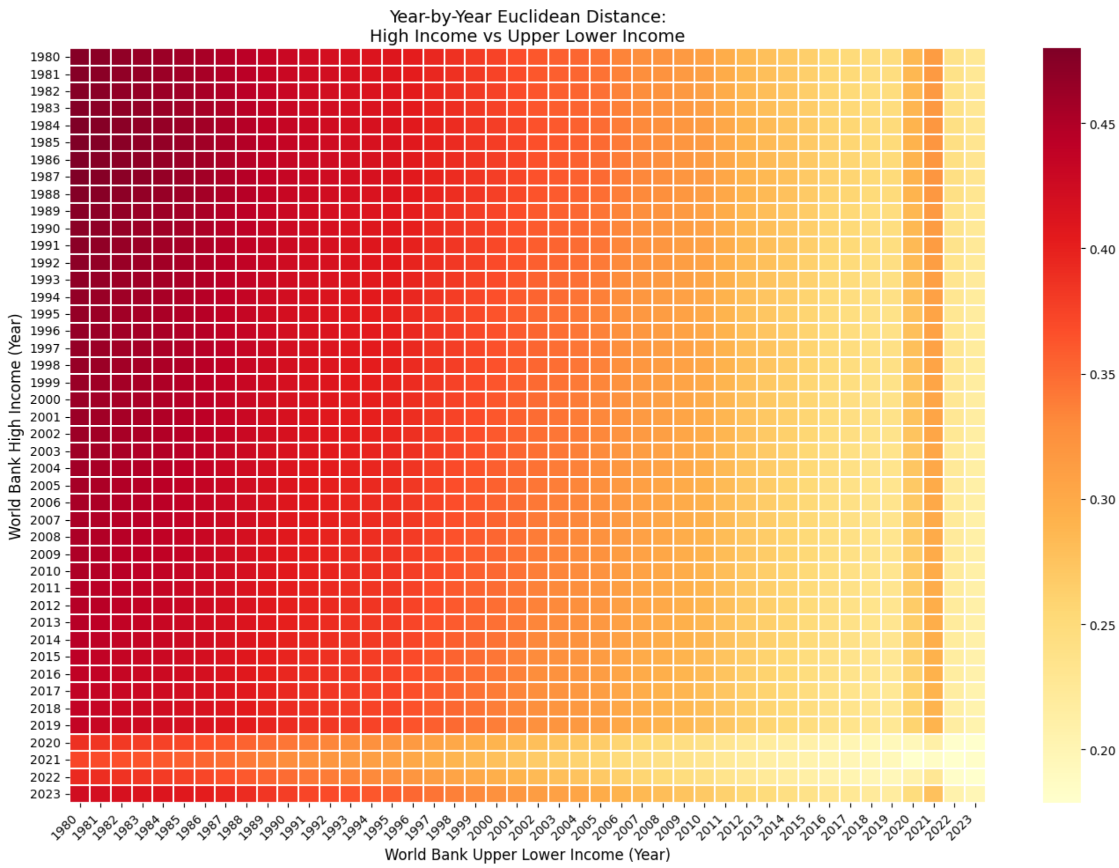
[19]:
```python
df = death_pivot.copy()

feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
↪'year']]
```

```
hi = df[df['income_class'] == 'World Bank High Income']
umi = df[df['income_class'] == 'World Bank Upper Middle Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

dist_matrix = pairwise_distances(
    hi_by_year.loc[years],
    umi_by_year.loc[years],
    metric='euclidean'
)

df_year_dist = pd.DataFrame(dist_matrix, index=years, columns=years)

fig, ax = plt.subplots(figsize=(14, 10))

sns.heatmap(
    df_year_dist,
    annot=False,
    fmt='.3f',
    cmap='YlOrRd',
    ax=ax,
    linewidths=0.3,
    linecolor='white'
)

ax.set_xlabel('World Bank Upper Middle Income (Year)', fontsize=12)
ax.set_ylabel('World Bank High Income (Year)', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nHigh Income vs Upper Middle␣
 ↪Income', fontsize=14)

plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```



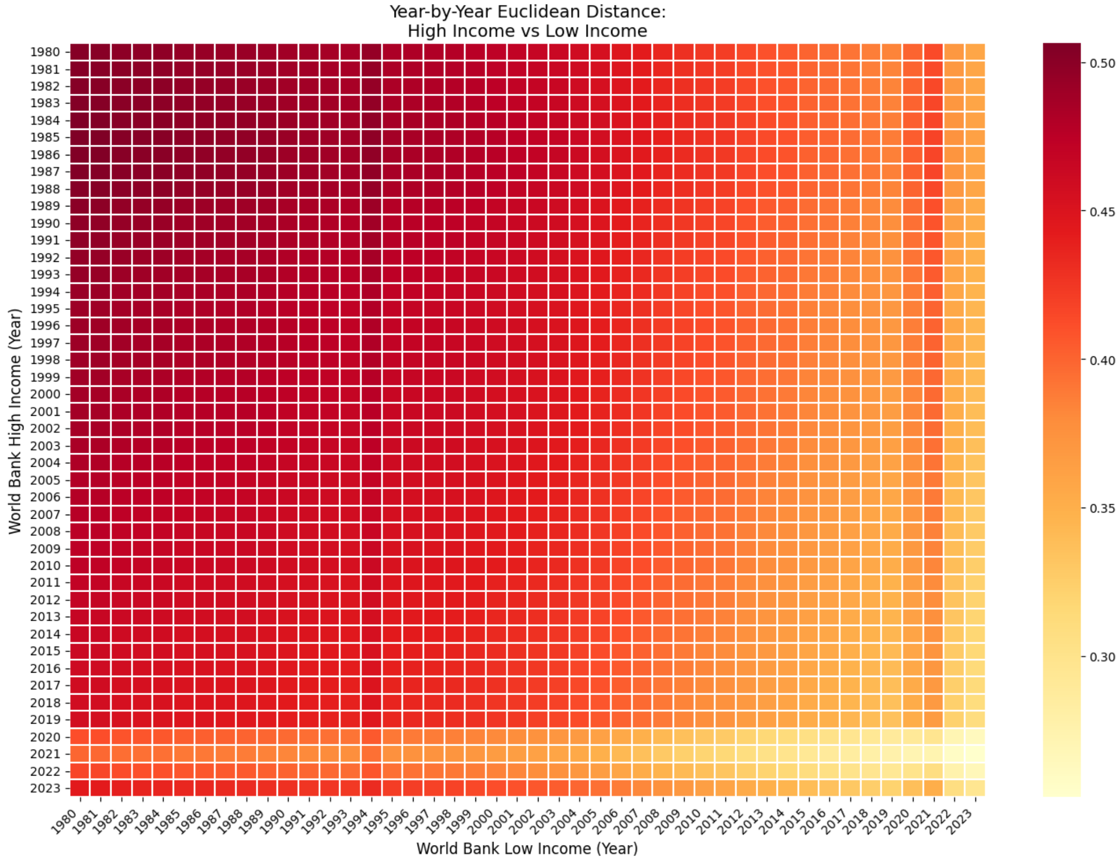
[20]:
```python
df = death_pivot.copy()

feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
↪'year']]

hi = df[df['income_class'] == 'World Bank High Income']
umi = df[df['income_class'] == 'World Bank Lower Middle Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

dist_matrix = pairwise_distances(
    hi_by_year.loc[years],
    umi_by_year.loc[years],
    metric='euclidean'
)

df_year_dist = pd.DataFrame(dist_matrix, index=years, columns=years)
```

```python
fig, ax = plt.subplots(figsize=(14, 10))

sns.heatmap(
    df_year_dist,
    annot=False,
    fmt='.3f',
    cmap='YlOrRd',
    ax=ax,
    linewidths=0.3,
    linecolor='white'
)

ax.set_xlabel('World Bank Upper Lower Income (Year)', fontsize=12)
ax.set_ylabel('World Bank High Income (Year)', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nHigh Income vs Upper Lower␣
 ↪Income', fontsize=14)

plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```



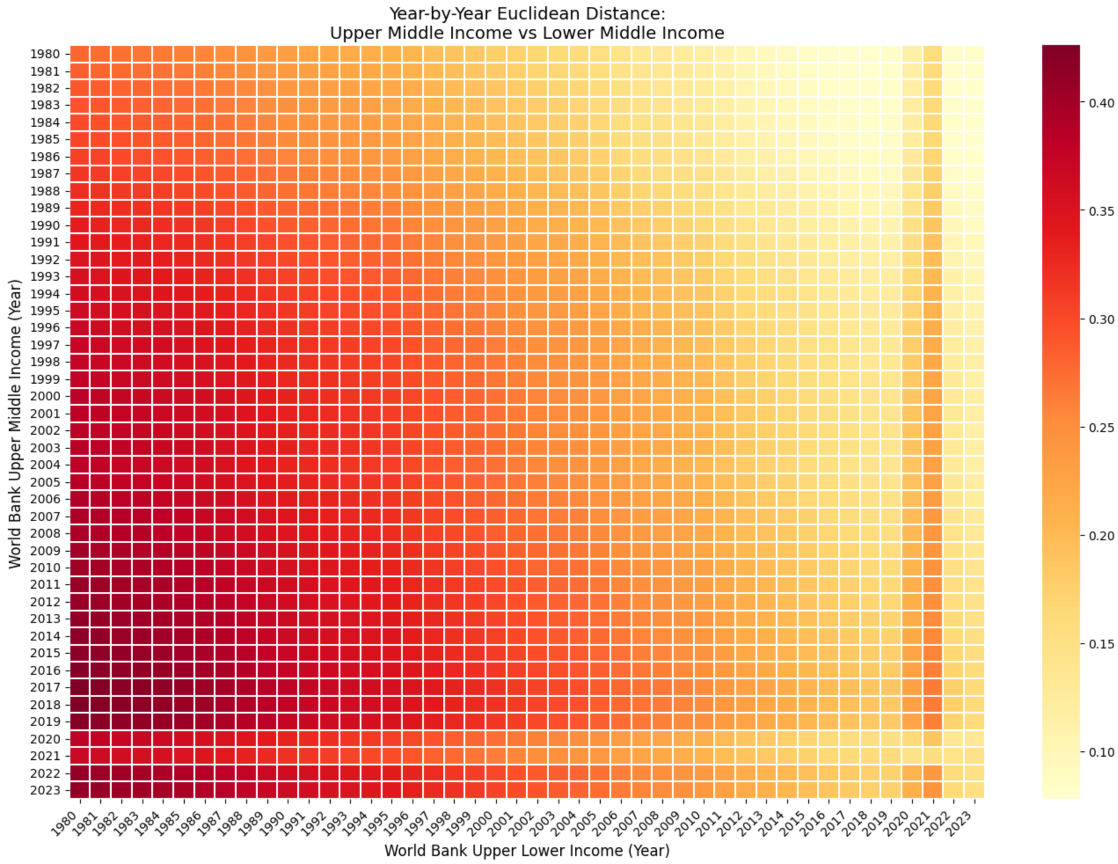
[21]:

```python
df = death_pivot.copy()

feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
↪'year']]

hi = df[df['income_class'] == 'World Bank High Income']
umi = df[df['income_class'] == 'World Bank Low Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

dist_matrix = pairwise_distances(
    hi_by_year.loc[years],
    umi_by_year.loc[years],
    metric='euclidean'
)

df_year_dist = pd.DataFrame(dist_matrix, index=years, columns=years)

fig, ax = plt.subplots(figsize=(14, 10))

sns.heatmap(
    df_year_dist,
    annot=False,
    fmt='.3f',
    cmap='YlOrRd',
    ax=ax,
    linewidths=0.3,
    linecolor='white'
)

ax.set_xlabel('World Bank Low Income (Year)', fontsize=12)
ax.set_ylabel('World Bank High Income (Year)', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nHigh Income vs Low Income',␣
↪fontsize=14)

plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```



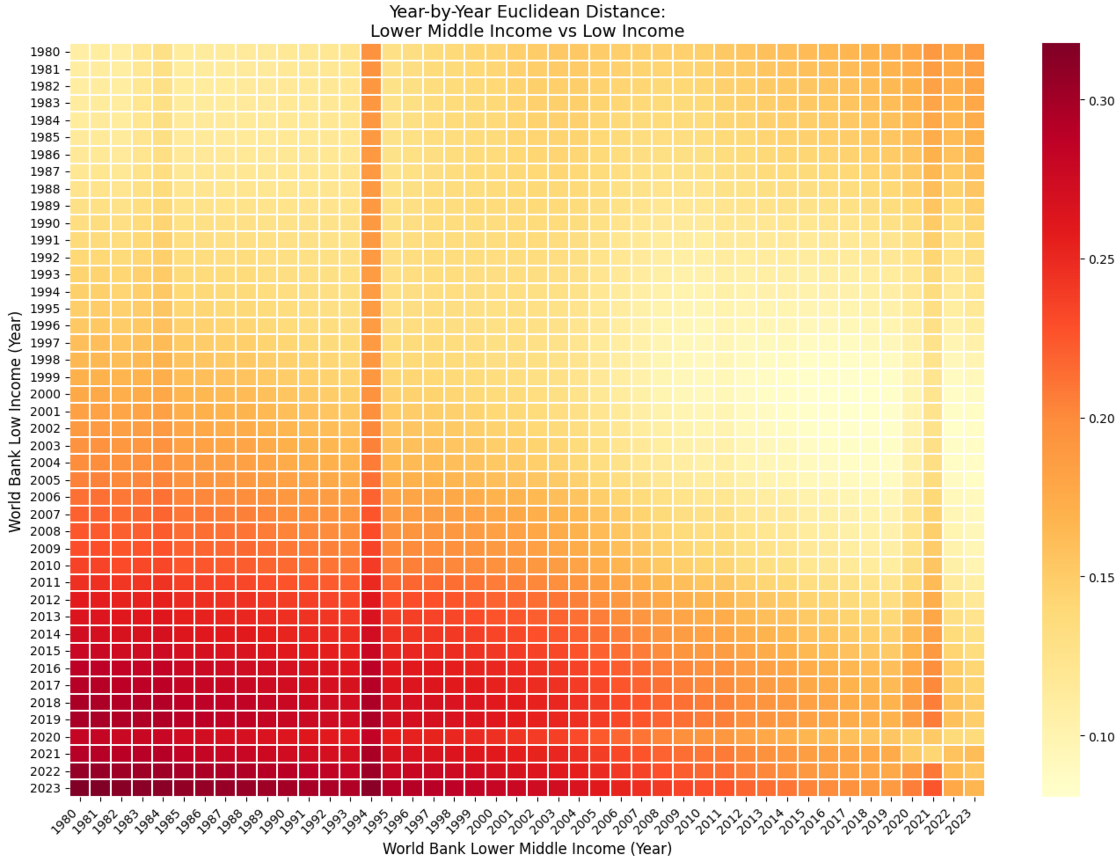

```
[22]: df = death_pivot.copy()

feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
 ↪'year']]

hi = df[df['income_class'] == 'World Bank Upper Middle Income']
umi = df[df['income_class'] == 'World Bank Lower Middle Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

dist_matrix = pairwise_distances(
    hi_by_year.loc[years],
    umi_by_year.loc[years],
    metric='euclidean'
)

df_year_dist = pd.DataFrame(dist_matrix, index=years, columns=years)
```

```
fig, ax = plt.subplots(figsize=(14, 10))

sns.heatmap(
    df_year_dist,
    annot=False,
    fmt='.3f',
    cmap='YlOrRd',
    ax=ax,
    linewidths=0.3,
    linecolor='white'
)

ax.set_xlabel('World Bank Upper Lower Income (Year)', fontsize=12)
ax.set_ylabel('World Bank Upper Middle Income (Year)', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nUpper Middle Income vs Lower␣
↪Middle Income', fontsize=14)

plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```



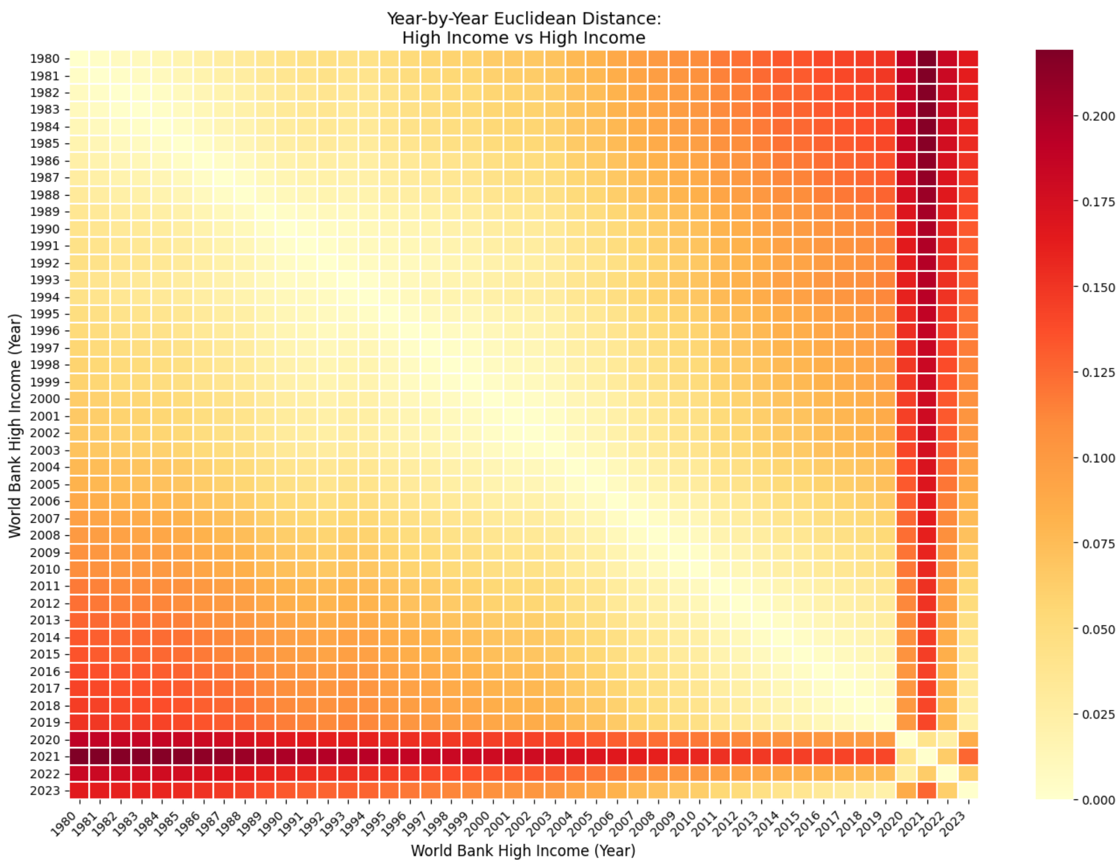
[23]:

```
df = death_pivot.copy()

feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
↪'year']]

hi = df[df['income_class'] == 'World Bank Lower Middle Income']
umi = df[df['income_class'] == 'World Bank Low Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

dist_matrix = pairwise_distances(
    hi_by_year.loc[years],
    umi_by_year.loc[years],
    metric='euclidean'
)

df_year_dist = pd.DataFrame(dist_matrix, index=years, columns=years)

fig, ax = plt.subplots(figsize=(14, 10))

sns.heatmap(
    df_year_dist,
    annot=False,
    fmt='.3f',
    cmap='YlOrRd',
    ax=ax,
    linewidths=0.3,
    linecolor='white'
)

ax.set_xlabel('World Bank Lower Middle Income (Year)', fontsize=12)
ax.set_ylabel('World Bank Low Income (Year)', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nLower Middle Income vs Low␣
↪Income', fontsize=14)

plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```




```
[24]: df = death_pivot.copy()

feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
 ↪'year']]

hi = df[df['income_class'] == 'World Bank High Income']
umi = df[df['income_class'] == 'World Bank High Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

dist_matrix = pairwise_distances(
    hi_by_year.loc[years],
    umi_by_year.loc[years],
    metric='euclidean'
)

df_year_dist = pd.DataFrame(dist_matrix, index=years, columns=years)
```

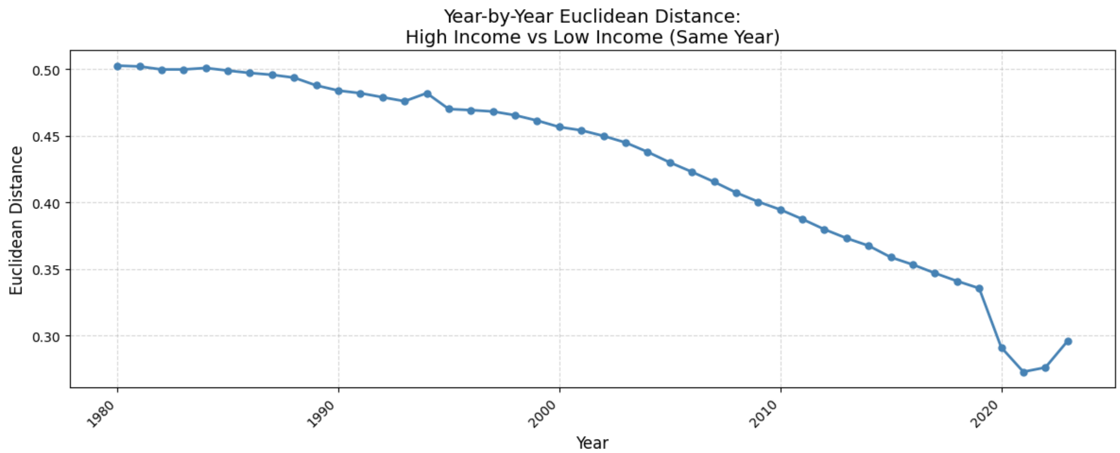
```python
fig, ax = plt.subplots(figsize=(14, 10))

sns.heatmap(
    df_year_dist,
    annot=False,
    fmt='.3f',
    cmap='YlOrRd',
    ax=ax,
    linewidths=0.3,
    linecolor='white'
)

ax.set_xlabel('World Bank High Income (Year)', fontsize=12)
ax.set_ylabel('World Bank High Income (Year)', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nHigh Income vs High Income', fontsize=14)

plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
```

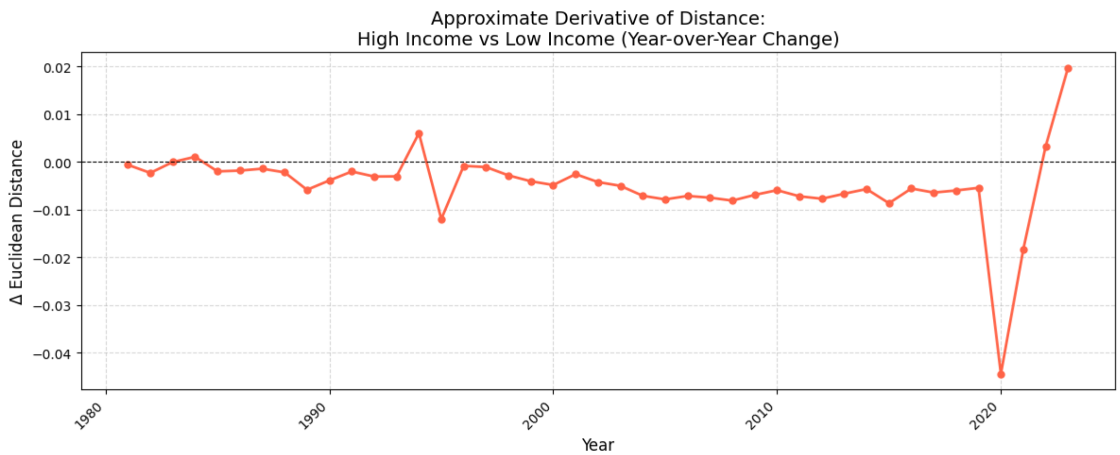


[25]:

```
df = death_pivot.copy()
feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
↪'year']]

hi = df[df['income_class'] == 'World Bank High Income']
umi = df[df['income_class'] == 'World Bank Low Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

diagonal_distances = [
    np.linalg.norm(hi_by_year.loc[y].values - umi_by_year.loc[y].values)
    for y in years
]

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(years, diagonal_distances, marker='o', linewidth=2, markersize=5,␣
↪color='steelblue')
ax.set_xlabel('Year', fontsize=12)
ax.set_ylabel('Euclidean Distance', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nHigh Income vs Low Income (Same␣
↪Year)', fontsize=14)
ax.grid(True, linestyle='--', alpha=0.5)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

years_mid = years[1:]
deltas = [diagonal_distances[i+1] - diagonal_distances[i] for i in␣
↪range(len(diagonal_distances)-1)]

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(years_mid, deltas, marker='o', linewidth=2, markersize=5,␣
↪color='tomato')
ax.axhline(0, color='black', linewidth=0.8, linestyle='--')
ax.set_xlabel('Year', fontsize=12)
ax.set_ylabel('Δ Euclidean Distance', fontsize=12)
ax.set_title('Approximate Derivative of Distance:\nHigh Income vs Low Income␣
↪(Year-over-Year Change)', fontsize=14)
ax.grid(True, linestyle='--', alpha=0.5)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
```

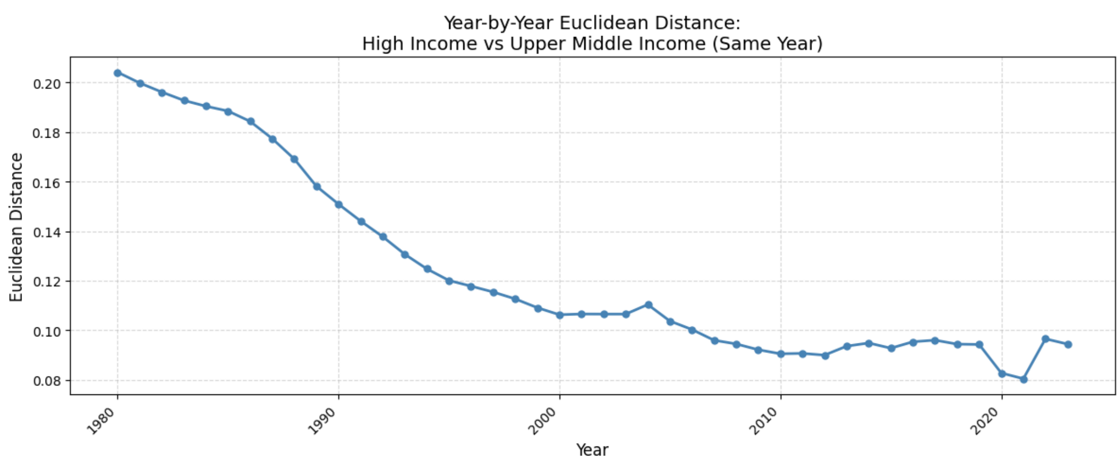
```
plt.show()
```

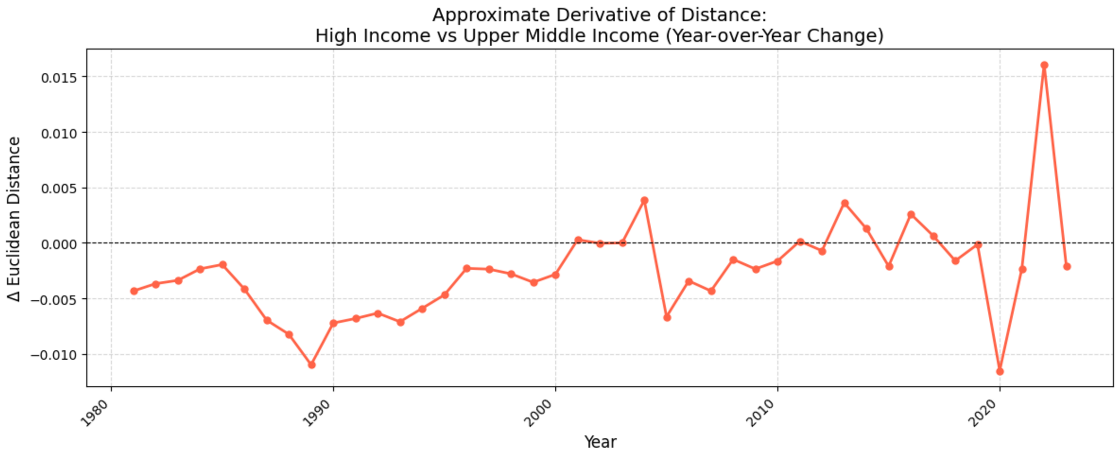




[26]:
```
df = death_pivot.copy()
feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
↪'year']]

hi = df[df['income_class'] == 'World Bank High Income']
umi = df[df['income_class'] == 'World Bank Upper Middle Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

diagonal_distances = [
```

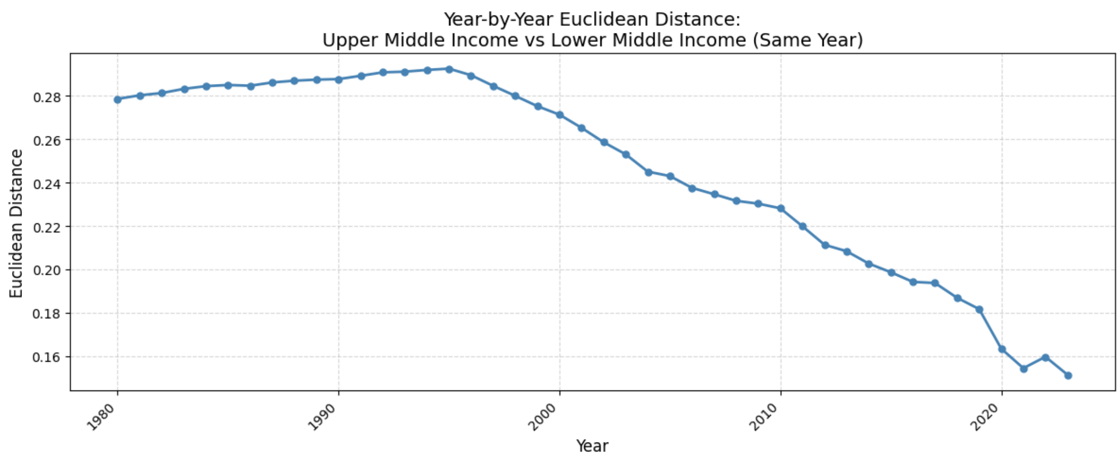
```
    np.linalg.norm(hi_by_year.loc[y].values - umi_by_year.loc[y].values)
    for y in years
]

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(years, diagonal_distances, marker='o', linewidth=2, markersize=5,
 ↪color='steelblue')
ax.set_xlabel('Year', fontsize=12)
ax.set_ylabel('Euclidean Distance', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nHigh Income vs Upper Middle
 ↪Income (Same Year)', fontsize=14)
ax.grid(True, linestyle='--', alpha=0.5)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

years_mid = years[1:]
deltas = [diagonal_distances[i+1] - diagonal_distances[i] for i in
 ↪range(len(diagonal_distances)-1)]

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(years_mid, deltas, marker='o', linewidth=2, markersize=5,
 ↪color='tomato')
ax.axhline(0, color='black', linewidth=0.8, linestyle='--')
ax.set_xlabel('Year', fontsize=12)
ax.set_ylabel('Δ Euclidean Distance', fontsize=12)
ax.set_title('Approximate Derivative of Distance:\nHigh Income vs Upper Middle
 ↪Income (Year-over-Year Change)', fontsize=14)
ax.grid(True, linestyle='--', alpha=0.5)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

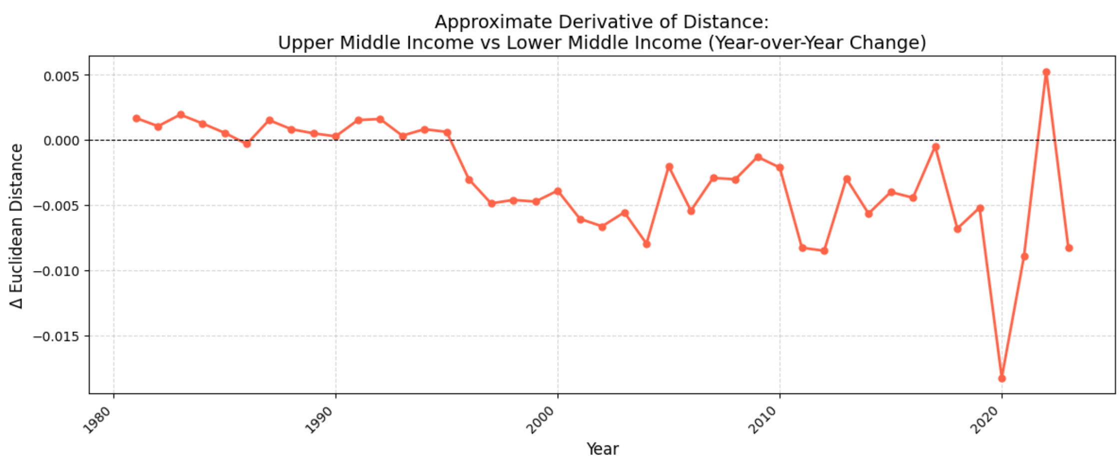




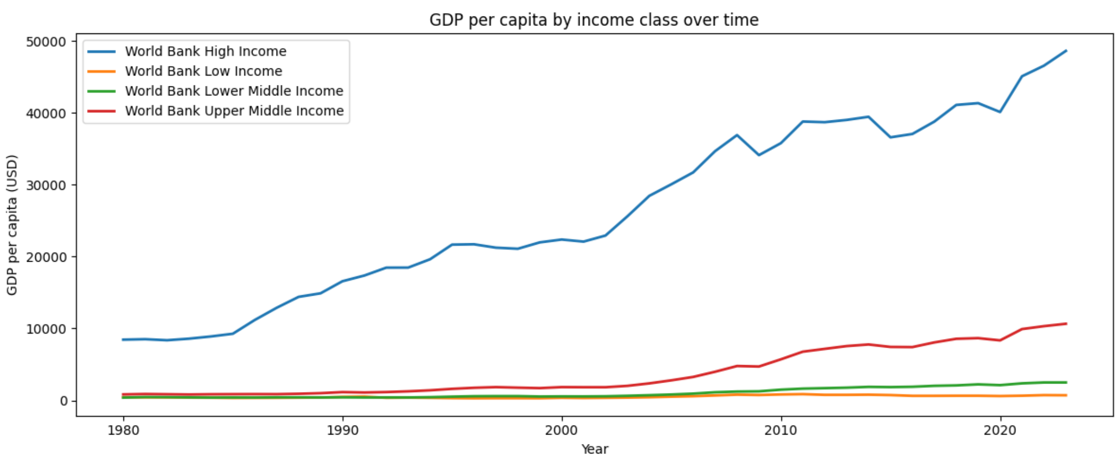
[27]:
```
df = death_pivot.copy()
feature_cols = [c for c in df.columns if c not in ['cause', 'income_class',␣
↪'year']]

hi = df[df['income_class'] == 'World Bank Upper Middle Income']
umi = df[df['income_class'] == 'World Bank Lower Middle Income']

years = sorted(set(hi['year'].unique()) & set(umi['year'].unique()))

hi_by_year  = hi.groupby('year')[feature_cols].mean()
umi_by_year = umi.groupby('year')[feature_cols].mean()

diagonal_distances = [
    np.linalg.norm(hi_by_year.loc[y].values - umi_by_year.loc[y].values)
    for y in years
]

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(years, diagonal_distances, marker='o', linewidth=2, markersize=5,␣
↪color='steelblue')
ax.set_xlabel('Year', fontsize=12)
ax.set_ylabel('Euclidean Distance', fontsize=12)
ax.set_title('Year-by-Year Euclidean Distance:\nUpper Middle Income vs Lower␣
↪Middle Income (Same Year)', fontsize=14)
ax.grid(True, linestyle='--', alpha=0.5)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
years_mid = years[1:]
deltas = [diagonal_distances[i+1] - diagonal_distances[i] for i in␣
 ↪range(len(diagonal_distances)-1)]

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(years_mid, deltas, marker='o', linewidth=2, markersize=5,␣
 ↪color='tomato')
ax.axhline(0, color='black', linewidth=0.8, linestyle='--')
ax.set_xlabel('Year', fontsize=12)
ax.set_ylabel('Δ Euclidean Distance', fontsize=12)
ax.set_title('Approximate Derivative of Distance:\nUpper Middle Income vs Lower␣
 ↪Middle Income (Year-over-Year Change)', fontsize=14)
ax.grid(True, linestyle='--', alpha=0.5)
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```





[28]:
```python
gdp_by_class = (
    df_income
    .groupby(['income_class', 'year'])['gdp_per_capita_usd']
    .first()
    .reset_index()
    .sort_values('year')
)

years = sorted(gdp_by_class['year'].unique().tolist())

plt.figure(figsize=(12, 5))
for cls, grp in gdp_by_class.groupby('income_class'):
    plt.plot(grp['year'], grp['gdp_per_capita_usd'], label=cls, linewidth=2)

plt.xlabel('Year')
plt.ylabel('GDP per capita (USD)')
plt.title('GDP per capita by income class over time')
plt.legend()
plt.tight_layout()
plt.show()
```



[29]:
```python
low = gdp_by_class[
    (gdp_by_class['income_class'] == 'World Bank High Income')
].copy()

low.sort_values("year", ascending=True).head(40)
```

[29]:

	income_class	year	gdp_per_capita_usd
0	World Bank High Income	1980	8449.5621

1	World Bank High Income	1981	8511.1153
2	World Bank High Income	1982	8366.5297
3	World Bank High Income	1983	8585.1327
4	World Bank High Income	1984	8892.1678
5	World Bank High Income	1985	9269.5947
6	World Bank High Income	1986	11184.5378
7	World Bank High Income	1987	12880.4034
8	World Bank High Income	1988	14398.6650
9	World Bank High Income	1989	14891.7846
10	World Bank High Income	1990	16573.1449
11	World Bank High Income	1991	17366.3731
12	World Bank High Income	1992	18462.7676
13	World Bank High Income	1993	18470.3300
14	World Bank High Income	1994	19634.0843
15	World Bank High Income	1995	21664.2004
16	World Bank High Income	1996	21713.2097
17	World Bank High Income	1997	21240.4885
18	World Bank High Income	1998	21090.4318
19	World Bank High Income	1999	21976.6115
20	World Bank High Income	2000	22374.3255
21	World Bank High Income	2001	22083.4202
22	World Bank High Income	2002	22929.7567
23	World Bank High Income	2003	25606.6766
24	World Bank High Income	2004	28451.5065
25	World Bank High Income	2005	30051.3460
26	World Bank High Income	2006	31710.7967
27	World Bank High Income	2007	34666.0452
28	World Bank High Income	2008	36900.1406
29	World Bank High Income	2009	34111.8446
30	World Bank High Income	2010	35778.2487
31	World Bank High Income	2011	38785.8817
32	World Bank High Income	2012	38693.8455
33	World Bank High Income	2013	39008.3249
34	World Bank High Income	2014	39447.7093
35	World Bank High Income	2015	36586.4347
36	World Bank High Income	2016	37057.8097
37	World Bank High Income	2017	38776.5616
38	World Bank High Income	2018	41094.8561
39	World Bank High Income	2019	41342.2066